# Minor Project Report

*August – December 2023*

Closed Wing
analysis

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Under the Guidance of
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# DECLARATION

We hereby declare that the minor project work titled “Analysis of Closed Wing Design” is our own work carried out at Our Institute as part of the minor project for the B.Tech. Aerospace Engineering, at Punjab Engineering College. This work was done under the guidance of Prof. Tushar Siag.

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Date ____________________

Certified by :

Prof. Tushar siag

# Acknowledgement

Our sincere gratitude to our mentor , Prof.Tushar Siag for their constant support throughout the whole project. We would also like to thank Mr. Pradeep for her guidance and enhancement of our experiments .

# 1. Introduction

The Blériot III, which emerged in 1906 through the collaborative efforts of Louis Blériot and Gabriel Voisin, stands as a pioneering closed-wing aircraft featuring a tandem configuration. Fast forward to 1924, and the theoretical groundwork for the box wing's advantages was laid by Prandtl. He underscored the significance of a lifting system with minimal induced drag, specifying conditions such as identical lift distribution along both upper and lower wings.

Frediani and Montanari entered the scene in 1999, providing a precise solution that corroborated Prandtl's assertions. Their findings revealed a noteworthy 20%-30% reduction in induced drag for box-wing configurations, particularly when the gap/span ratio fell within the 10-20% range.

Taking a leap to 1976, Julian Wolkovitch secured a patent for the joined-wing configuration, a design that came to prominence in 1985 with the inaugural flight of Ligeti Stratos. This innovative design featured wings that not only provided essential lift but also formed a structural frame. The advantages associated with the joined-wing configuration are diverse and impactful. They include a reduction in the weight of the wing structure, heightened structural stiffness, and a lower induced drag when compared to equivalent cantilever wings.

The benefits extend further to encompass improved transonic pressure distribution, a commendable glide ratio, diminished wetted surface, and the added advantages of direct control over both lift and side forces. These advantages, as highlighted, are substantiated by a wealth of independent analyses, thorough design investigations, and meticulous wind tunnel experiments.

This joined-wing configuration isn't limited in its applicability. It shows promise for various aircraft scales, from large-scale options capable of surpassing the payload capacity of the A380 while utilizing existing airport infrastructure to smaller aircraft and unmanned aerial vehicles. Additionally, there's a notable suggestion that the box wing, Prandtl plane, and other configurations are specific instances falling under the broader umbrella of joined wings.

# 2. Classification indicators

The following classification of closed wing aircraft has seven basic indicators. As is the case with all classifications of objects that are rapidly developing it can be improved and expanded, in order to be successfully applied for the conceptual design of aircraft.

## 2.1. Wing planform

Fig.2.1.1. Rhomboidal wing

Fig.2.1.2. Triangular wing

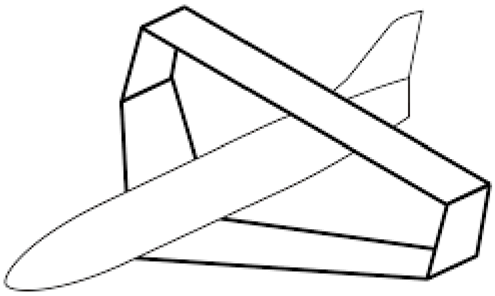


Fig.2.1.3. Straight wing

## 2.2. Wing shape when looked from the front

Fig.2.2.1. Cylindrical wing

Fig.2.2.2. Annular wing

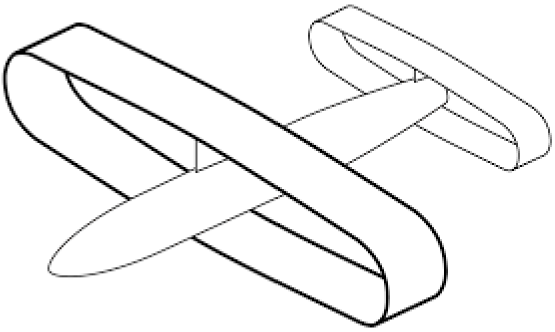


Fig.2.2.3. Elliptic wing

Fig.2.2.4. Box wing

Fig.2.2.5. Diamond-shape wing

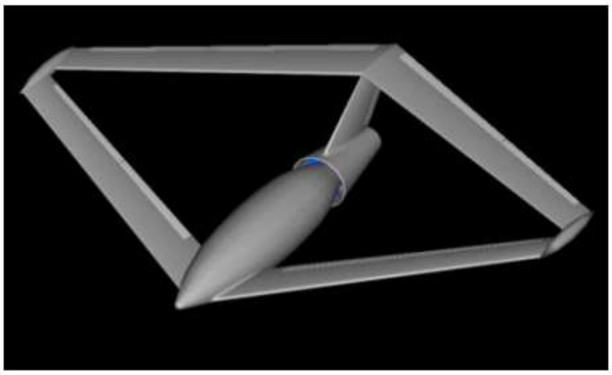


Fig.2.2.6. Trapezium-shaped wing

Fig.2.2.7. Flat wing

Fig.2.2.8. Joined wing with winglets

## 2.3. Wing shape when looked from the side

Fig.2.3.1. Front wing lower than the rear wing

Fig.2.3.2. Front wing higher than the rear wing

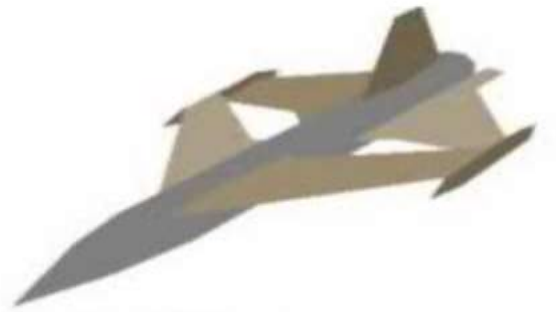


Fig.2.3.3. Front wing in the same plane with the rear one

## 2.4. Joint location

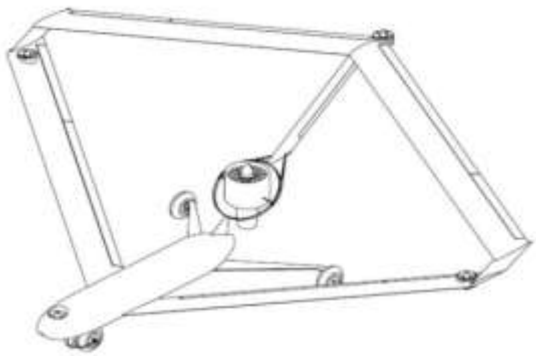
Fig.2.4.1. At the tips

Fig.2.4.2. Along the span

## 2.5. Takeoff and landing type

Fig.2.5.1. Conventional takeoff and landing airplanes

Fig.2.5.2. Vertical takeoff and landing airplanes



## 2.6. Flight control method

Fig.2.6.1. With control surfaces

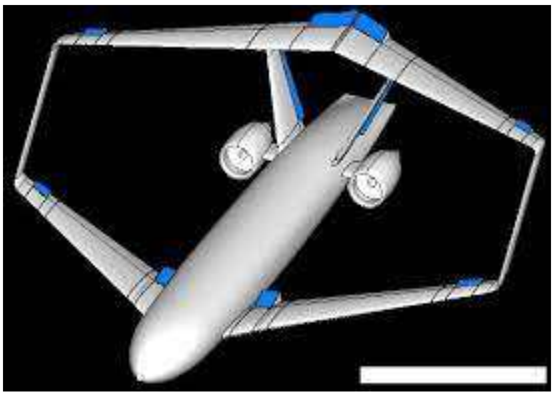


Fig.2.6.2. With control propulsors

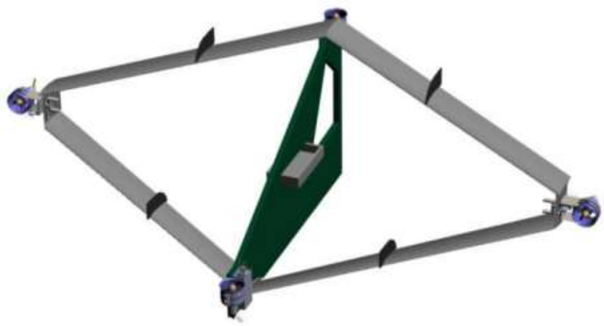
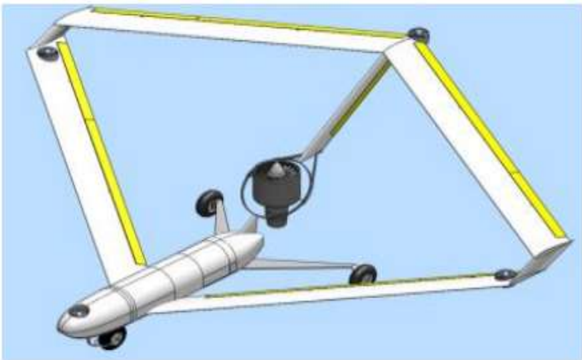


Fig.2.6.3. Combined - with control surfaces and propulsors



**2.7. Payload arrangement**

Fig.2.7.1. In a fuselage

Fig.2.7.2. In the wing

## 3. Discuss the advantages, disadvantages and prospects for closed wing concepts for commercial passenger aircraft

Aircraft designs are always looking to be improved. As our understanding of plane aerodynamics increases, humans have been able to create faster or more efficient aircraft. We will forever be seeking new designs or features for aircraft that provide the optimum level of lift for the plane, or the minimum level of drag. So far the majority of commercial aircraft have used a standard configuration of a fixed open wing design, but there are concepts being made for commercial aircraft with very different designs. This includes closed wing aircraft, and blended wing body aircraft. I am going to research the closed wing aircraft design for commercial aircraft, and look to see if this is a possible area of profitable design for the future.

The closed wing design is one where the wings extend from each side of the plane and the neither turn to reattach to the plane on the same side creating a box wing, or come back to connect with the wing from the other side of the fuselage creating one large wing surrounding the aircraft. Various shapes of closed wings are possible, including cylindrical wings, box wings and joined wings. This type of design, if developed and found possible to provide for successful commercial use, may be a large step for the progression of the efficiency of commercial aircraft.

The largest benefit found from using this design is the reduction in drag from the wings, with some designs of closed wing providing flight with up to 50% less drag [1]. At the wing tips of conventional open wing aircraft, we get wing tip vortices. These are circular patterns of rotating air that trail the aircraft, which are created by the wing as it produces lift. By using the closed wing design, we have no wing tip, which means that the amount of drag produced by the wing is reduced greatly. If an aircraft experiences lower drag, then this means it will be capable to fly further on the same amount of fuel. This is because less thrust is required to be produced from the engines to reach the same speed as a plane with greater drag. This improvement will allow for more commercial aircraft to reach long range destinations in one journey, that are greater distances than are currently reachable in one flight. Therefore with greater efficiency in time and fuel, this will mean flights would become significantly cheaper to run. The decrease in fuel consumption is particularly important in our current global situation of rising oil prices.

Another benefit of the design is that the aircraft would cause less disruption in the sky. Wingtip vortices can leave air spiralling down through the atmosphere for minutes if the plane is large enough. Currently two aircraft categorised as Heavy (140,000Kg or more) would have to remain at a separation of at least four Nautical miles as they come in to land [2]. This is because flying into the rotating air left from another plane could cause the following aircraft to spin or to experience extreme turbulence, and at low altitudes control cannot always be regained quickly enough to avoid a crash. Many crashes have occurred in the past due to aircraft flying into wing tip vortices left by another plane. By using the closed wing design and reducing the size of the wing tip vortices, aircraft could fly in closer proximity to one another. This allows for more direct routes to be taken, reducing diversions which need to be made to avoid the trails of other planes. This is increasingly useful in a world with an ever rising number of commercial flights being taken.

However, there are some who claim the design does not provide benefits. Dr. Ilan Kroo, Professor of Aeronautics and Astronautics at Stanford University, has said that as there is still a continuous, uniform disruption to the air as the aircraft flies, there is no significant improvement in reducing the amount of wake and drag that is created [3]. The only benefit he has described is a better ability of spreading loading in the wings, helping to spread the weight they carry more evenly.

An aircraft with a closed wing design would also need to be altered in order to create a stable structure. If the design was one where the wings reattach above the plane, the fuselage would need to have a reinforced point to allow the wings to be held in place, as there would be a greater force acting on the fuselage. The standard aluminium frame used on commercial aircraft at the moment would most likely be insufficient to support a closed wing. The structure of the wing would also vary from the current design. As the wing is a greater length, travelling out and then back to the plane again, the wings may need to be made stronger, in order to hold their shape effectively. Sturdier wings may mean a greater level of turbulence is experienced in the aircraft, and a development in materials may be required to provide a wing strong enough to hold its shape, which can also absorb vibrations allowing for smooth flight within the aircraft, which is essential for commercial travel. Both of these alterations, of the wing and reinforcing the fuselage, would be expensive to develop. This may be a reason for the reluctance to venture into this new area of aircraft design.

A serious disadvantage of making large differences in design is also public opinion. It is quite common for people to have a fear of flying, or to at least be uncomfortable on a flight. This is even though the aircraft used are the safest way to travel in the world and have been proving it, in cases such as the Boeing 747, for the past 40 years. If a new design of aircraft was to be introduced, one with a radical change to current designs, people may be reluctant to fly on them. If this was the case, the commercial airlines that are running the aircraft may see a decrease in ticket sales as passengers opt to use airlines still running the current designs. If this was found to be the case, sales of the aircraft would tumble, proving the aircraft to be a financial failure.

To conclude I believe that closed wing aircraft will become the standard design of aircraft, but it is a question of how long it will take. Currently there is an obvious reluctance for a large company such as Airbus or Boeing to take the first step into a large project on a closed wing design. The fear of failure in the project is holding them back, as the risk of the project being unsuccessful is deemed too high to take. The design shows potential of far greater efficiency in the sky, with reduced drag allowing for aircraft to cover much greater distance due to a reduction in fuel consumption. And with less of a footprint in the sky from wake turbulence, it allows for greater flexibility in travel paths, this would be a huge improvement for flight in an ever crowded airspace. But with the possibility of an insignificant improvement in drag reduction and a poor public reaction to the design, whether or not closed wing aircraft are the future of commercial air travel is yet to be discovered.

# 4. Approach for the verification of Concept

We have gone through the various research papers, journals and article which us methodology that mostly opted and accepted by the experts is producing comparison of traditional wing design with subject wing design through mathematical modelling in multiple phases and verified that by the experiment in similar phases. This reduces the error deviation or tolerance in final result as mentioned in one of the article.

To achieve the same we have run the static loading simulation on wing via FEM. And CFD for aerodynamic characterisation of wings. All this simulations are iterated in the different conditions and parameters and the various factors like lift coefficient , drag coefficient and flexure stiffness contrast the capabilities if the reference and subject object.

The design optimization was carried out with a Risø in-house multi disciplinary optimization tool. Wind tunnel testing was performed for Risø-B1-18 and Risø-B1-24 in the wind tunnel, pec, at a Re = 1.63106. For both airfoils the predicted target performance were achieved.

Verification of Sliding Ratio (Cl/Cd) of Airfoil through CFD Analysis . And investigated the aerodynamic behaviour of RAE2822 in ground effect with FVM (finite volume method) based on the averaged Navier-Stokes equations. The performance of many eddy-viscosity turbulence models were assessed by comparing with the existing experimental data of NACA-4412 in ground effect. Realizable k-epsilon model showed high capability of predicting the characteristics of flow. In the study of this ground effect on airfoil RAE-2822, high lift/drag ratio can be achieved in medium AOA (angle of attack). To gain the increased lift with less drag through wing, the aerodynamic behavior of profile of the wing must be improved. The key parameters are the lift & drag coefficient to analyze the wing performance. To gain the largest range from the aircraft the maximum lift/drag ratio is required. (Talukder et al., 2016) performed a comparative analysis for aerodynamic performance of NREL S819 and S821 airfoils based on finite volume approach using a CFD method. Changing the angle of attack and air speed, different aerodynamic parameters such as lift coefficient, drag coefficient and pressure distribution over the airfoils were determined computationally.

The results from computations were confirmed experimentally by testing the airfoils foam models in a wind tunnel subsonic open circuit blower type. The comparison with the experimental data indicates that the CFD approach applied in this investigation can precisely predict the aerodynamic behavior of the wind-turbine wings. (Tenguria et al., 2017) performed CFD analysis of a wing as well as airfoil of HAWT using k-ω SST model. In this study, NACA 64A-204 airfoil profile was chosen for the modeling and then performs wing analysis. The lift & drag forces were find out for the wing at various AOA (angle of attack). The length of wing was taken 38.98 m. Results obtained from simulation were varified with the experimental work found in literature.

(Patil et al., 2015) studied the drag and lift forces wing at various Reynolds number and AOA. In this work NACA0012 airfoil profile was taken for analysis of wing. The drag and lift forces were find out by CFD analysis at various AOA from 0 degree to 80 degree for the Reynolds number in the range from 10000 to 800000. The validation of this work was done by comparing the result obtained from experimental results obtained by Sandia National Laboratories (SNL). It was concluded that result obtained by CFD analysis are very close with results published at SNL. Several computational techniques were applied, including a viscous/inviscid interaction method and three Reynolds-averaged Navier-Stokes methods. (Sharma, 2016) investigated to find out the most appropriate design of airfoil for using in low speed aircrafts. The airfoil S819, S1223, S1223 and S8037 RTL was chosen for study. The CFD method was used for analysis of airfoils. The numerical simulation performed using ANSYS FLUENT for low speed and high-lift airfoil. The coefficient of moment and Lift/Drag ratio of the airfoils find out for the comparative analysis of airfoils. The S1223 RTL airfoil was selected as the most appropriate design for the Mach number from 0.10 to 0.30 and for the specified boundary conditions. BEMT (Blade element momentum theory) is widely used for prediction of aerodynamic performance of wind turbine. But the reliability of the airfoil data is a significant aspect for the accurate prediction of power and aerodynamic forces. Mostly 2D wind tunnel tests of airfoils are done with constant span to establish the airfoil characteristics data used in BEM codes. Due to three dimensional effects, a BEM code using airfoil data received from two dimensional wind tunnel tests will not yield the correct loading and power. Consequently, two dimensional airfoil data have to be corrected before using in a BEM code. (Yang et al., 2014) considered the MEXICO rotor (Model Experiments in Controlled Conditions rotor) where airfoil data are extracted from CFD results. The comparison shows that the re-calculated forces by using airfoil data extracted from CFD have good agreements with the experiment.

In present work CFD analysis is carried out for airfoil NACA 63(4)-221 for redesigning the wing of RRB V27-225 kW HAWT. In this work the viscous spalart allmaras model is used. Spalart-Allmaras model is a one equation model which solves a transport equation for a viscosity-like variable, referred to as the Spalart-Allmaras variable. In its original form, the model is effectively a low-Reynolds number model. The Spalart–Allmaras model was developed for aerodynamic flows. The computational effort is lower compared to the commonly used two-equation models. This model is preferable in large part due to its robustness.

## III. Governing Equations

The continuity equation for the two dimensional, steady and incompressible flow is

$$\nabla.(\rho V) = \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0$$

For viscous flow in x direction the momentum equation is

$$\rho \frac{Du}{Dt} = \frac{\partial \rho}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x \qquad (2)$$

Where due to characteristics of the two dimensional flow in continuity equation the term $\frac{\partial(\rho w)}{\partial z}$ and in momentum equation, $\frac{\partial \tau_{zx}}{\partial z}$ drop out. In all simulations a standard k- ω SST model has been used for turbulent viscosity.

In equation (1) and (2)

$\rho$ = Density of fluid,
V = Velocity vector,
$\rho V$ = Mass flux,
$\nabla$ = Vector operator,
$\nabla.(\rho V)$ = Divergence of $Pv$,
$\rho u$, $\rho v$ = Rate of mass entering in x, y direction, respectively
$\tau$ = Shear stress,
$f_x$ = External force,
$\rho \frac{Du}{Dt}$ = Substantial time derivative of velocity,
u =Velocity vector in x direction

IV. BOUNDARY CONDITION AND GEOMETRY

In present analysis, an airfoil from the 6 series of NACA laminar wing section family is used. The airfoil maximum relative thickness is 21%, which is situated at 35% of the chord length. The Reynolds number taken for the simulation in the range of x105 and x106 and turbulence intensity is set at 10%. A turbulent flow solver is used in ANSYS Fluent, where spalart allmaras model is used. Calculation was performed for the —linear‖ region, i.e. for angles of attack (AOA) ranging from 4° to 8°, because of greater reliability of both computed and experimental values in this region. The selected airfoil profile has 50 no. vertices and it is created in ANSYS GUI with two edges upper and lower. The mesh is generated in ANSYS workbench and then boundary conditions are applied using ANSYS Fluent. Fig. 1 is showing the airfoil profile of NACA 63(4)-221. Once the airfoil edges were created then boundary layers are generated around the airfoil. Meshed around airfoil at different angle of attack have shown from Fig. 2 to Fig. 5. The mesh generated is uniformly distributed around airfoil for accurate prediction.

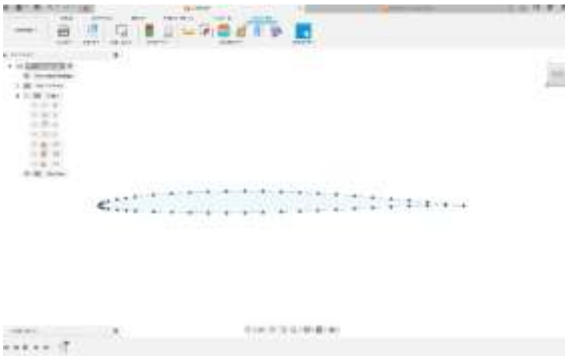
## V. CFD ANALYSIS OF AIRFOIL 64A-204 FOR 14 M

HAWT WING USING ANSYS FLUENT

A. Preprocecessing

1) Preparation of CAD Model

2D CAD model of NACA airfoil 64A-204 is generated using ANSYS design modeler. Fig. 1 shows modelling of NACA airfoil 64A-201. Coordinates are shown in Table I named airfoil specification of NACA 64A-201[4].



| Upper surface | | Lower surface | |
|---|---|---|---|
| Station | Ordinate | Station | Ordinate |
| 0.00000 | 0.0000 | 0.00000 | 0.00000 |
| 0.00367 | 0.01627 | 0.00633 | -0.01527 |
| 0.00600 | 0.02001 | 0.00900 | -0.01861 |
| 0.01075 | 0.02628 | 0.01425 | -0.02414 |
| 0.02292 | 0.03757 | 0.02708 | -0.03385 |
| 0.04763 | 0.05375 | 0.05237 | -0.04743 |
| 0.07253 | 0.06601 | 0.07747 | -0.05753 |
| 0.09753 | 0.07593 | 0.10247 | -0.06559 |
| 0.14767 | 0.09111 | 0.15233 | -0.07765 |
| 0.19792 | 0.10204 | 0.20208 | -0.08612 |
| 0.24824 | 0.10946 | 0.25176 | -0.09156 |
| 0.29860 | 0.11383 | 0.30140 | -0.09439 |
| 0.34897 | 0.11529 | 0.35103 | -0.09469 |
| 0.39934 | 0.11369 | 0.40066 | -0.09227 |
| 0.44969 | 0.10949 | 0.45031 | -0.08759 |
| 0.50000 | 0.10309 | 0.50000 | -0.08103 |
| 0.55027 | 0.09485 | 0.54973 | -0.07295 |
| 0.60048 | 0.08512 | 0.59952 | -0.06370 |
| 0.65063 | 0.07426 | 0.64937 | -0.05366 |
| 0.70071 | 0.06262 | 0.69929 | -0.04318 |
| 0.75073 | 0.05054 | 0.74927 | -0.03264 |
| 0.80067 | 0.03849 | 0.79933 | -0.02257 |
| 0.85056 | 0.02693 | 0.84944 | -0.01347 |
| 0.90039 | 0.01629 | 0.89961 | -0.00595 |
| 0.95018 | 0.00708 | 0.94982 | -0.00076 |
| 1.00000 | 0.00000 | 1.00000 | 0.00000 |
| LE Radius: 0.0265, slope of radius through LE 0.0842 | | | |

2) Mesh Generation

Generate the mesh of airfoil in the Ansys mesh software. (Fig. 2 to Fig. 6 at different angle of attack). Table II shows variables taken for meshing around the airfoil.

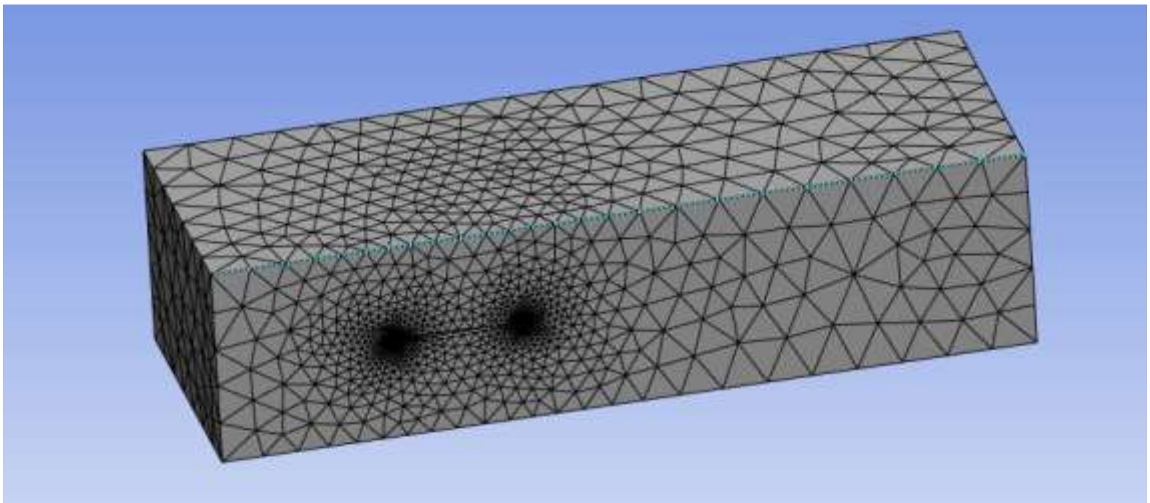


Figure . 4.1 . meshing on surface

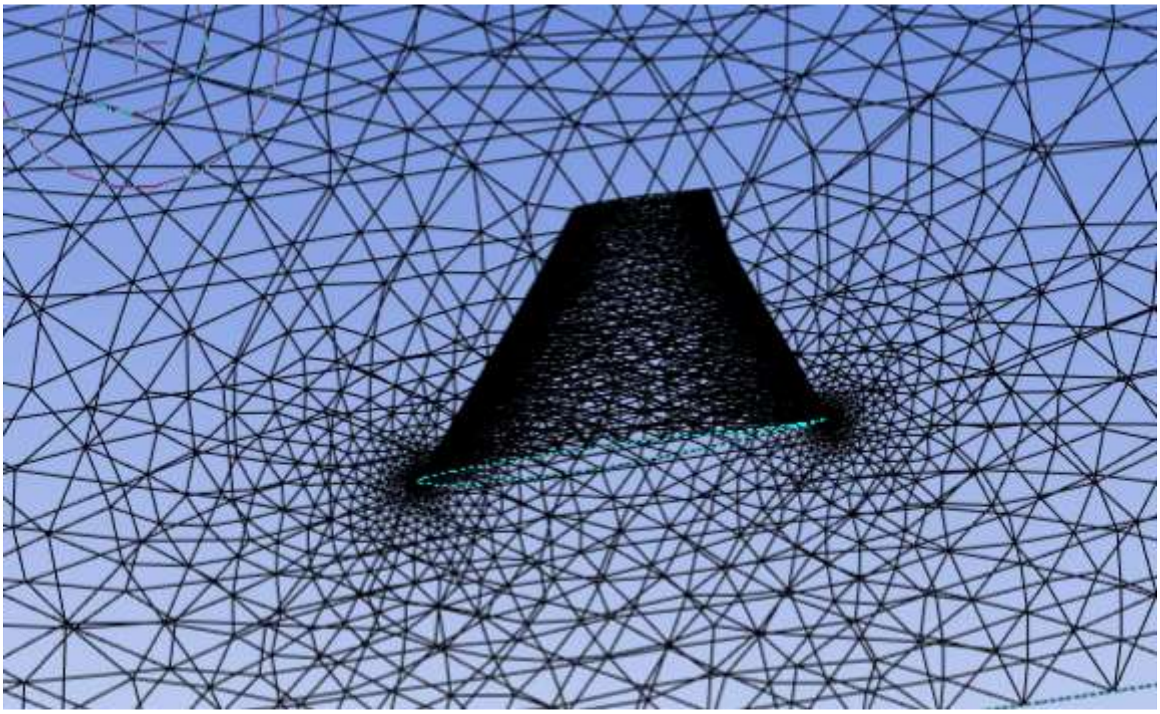


Figure 2.2 . meshing in volume

Table II. Variables for meshing around airfoil

| | |
|---|---|
| **Mesh Type** | Quadrilateral |
| **No. of Nodes** | 90985 |
| **No. of Element** | 90259 |

3) Fluent Setup

After mesh generation define the following setup criteria in the ANSYS fluent as shown in Table III. Pressure outlet conditions are shown in Table IV.

Table III. Setup criteria

| | |
|---|---|
| **Problem Type** | 2D |
| **Type of Solver** | Pressure- based solver |
| **Physical model** | Viscous spalart allmaras (1 equation) |
| **Material Property** | Flowing fluid is air |
| **Density of air** | 1.225 $kg/m^3$ |
| **Viscosity** | 1.7894e- 05 |

Table IV. Pressure outlet condition

| | |
|---|---|
| **Gauge pressure** | 0 Pa |
| **Turbulent viscosity ratio** | 10 |

B. Solution

1) Step 1-Solution Initialization

In this step Initialization of the solution is conduct to get the initial solution for the problem.

2) Step 2- Run Solution

Run the solution by giving 500 no of iteration for solution to converge. Following solution method is used for achieving solution (Table V).

Table V. Solution Method details

| | |
|---|---|
| **Pressure velocity coupling Scheme** | COUPLED |
| **Pressure** | Second order upwind |
| **Momentum** | Second order upwind |
| **Modified turbulence viscosity** | First order |

C. Post Processing

Post processing is performed for viewing and interpretation of the result. The result can be viewed in various formats like graph, value, animation etc. From Fig. 7 to 30 shows Pressure distribution, Velocity distribution and Velocity vector for selected airfoil at different wind velocity and angle of attack = 6°. These CFD results were also find out for other angle of attack as mention in Table VI and VII. In these tables lift force L, drag force D, lift coefficient Cl, drag coefficient Cd and Cl/Cd ratio is shown. It is clear from these results that angle of attack 6° give highest lift coefficient and Cl/Cd ratio. Values are shown in Table 6 and 7 for different Reynolds number in the range of 105 and 106 separately. Correction factor = 0.4 multiplied to drag coefficient, to get corrected drag coefficient as explained by (Ramanujam et al., 2016).

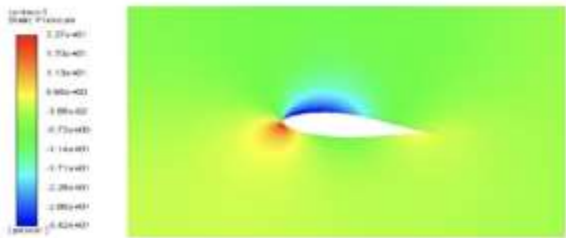


Fig. 7. Pressure distribution around Airfoil-NACA 63(4)-221 at AOA=6°and v=6 m/s

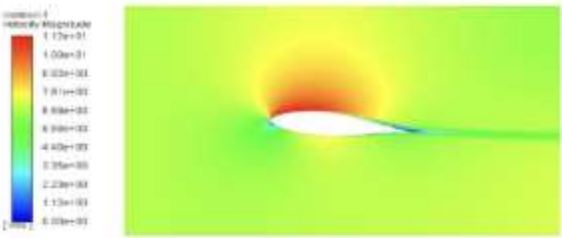


Fig. 11. Velocity distribution around Airfoil-NACA 63(4)-221 at AOA=6°and v=7 m/s

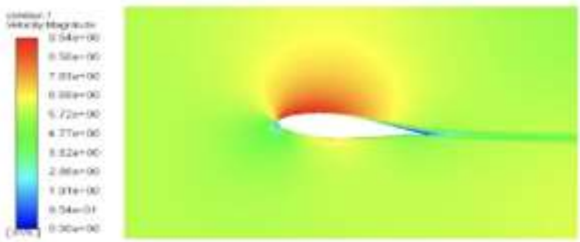


Fig. 8. Velocity distribution around Airfoil-NACA 63(4)-221 at AOA=6°and v=6 m/s

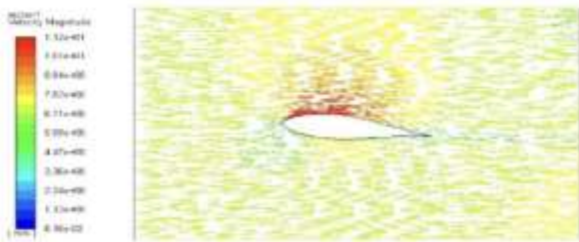


Fig. 12. Velocity vector around Airfoil-NACA 63(4)-221 at AOA=6° and v=7 m/s

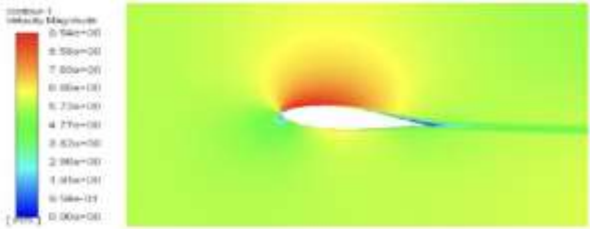


Fig. 9. Velocity vector around Airfoil-NACA 63(4)-221
AOA=6°and v=6 m/s

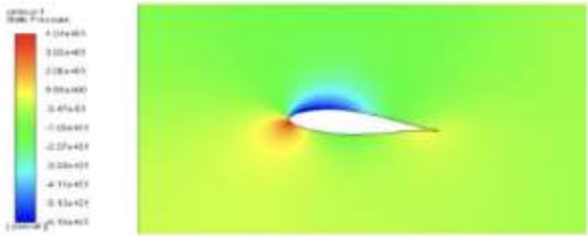


Fig. 13. Pressure distribution around Airfoil-NACA at
63(4)-221 at AOA=6° and v=8 m/s

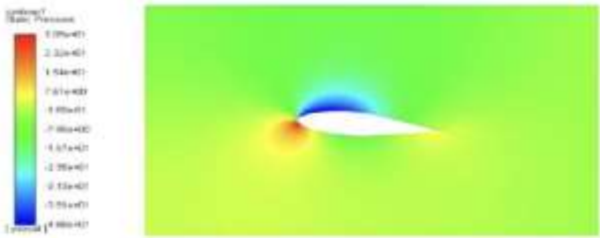


Fig. 10. Pressure distribution around Airfoil-NACA 63(4)-221

at AOA=6°and v=7 m/s

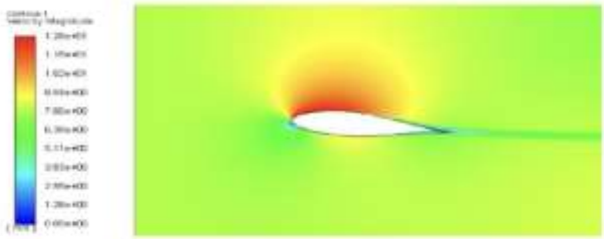


Fig. 14 Velocity distribution around Airfoil-NACA

63(4)-221 at AOA=6°and v=8 m/s

Fig. 15. Velocity vector around Airfoil-NACA 63(4)-221
at AOA=6°and v=8 m/s

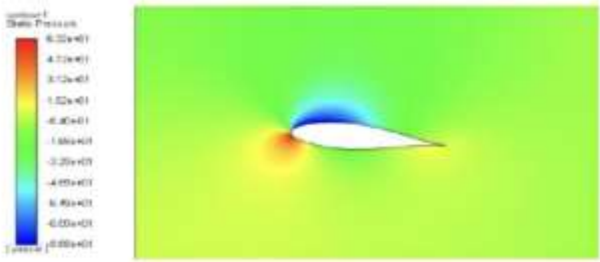


Fig. 19. Pressure distribution around Airfoil-
NACA 63(4)-221 at AOA=6°and v=10 m/s

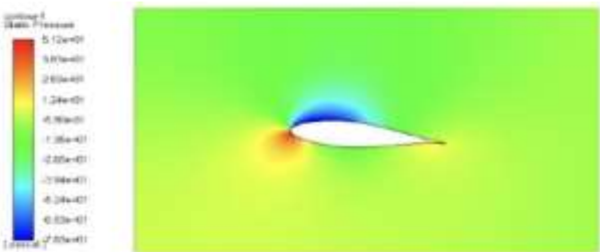


Fig. 16. Pressure distribution around Airfoil-NACA 63(4)-221
at AOA=6° and v=9 m/s

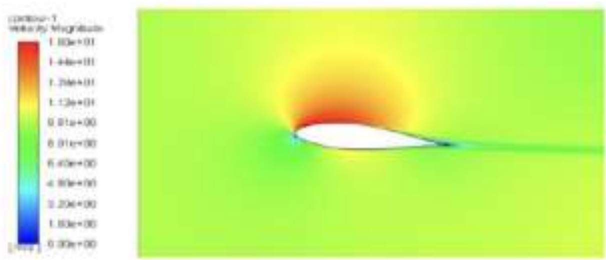


Fig. 20. Pressure distribution around Airfoil-
NACA 63(4)-221 at AOA=6° and v=10m/s

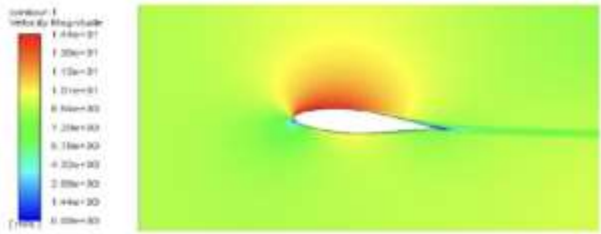


Fig. 17.Velocity distribution around Airfoil-NACA 63(4)-221 v=9 m/s

Fig. 21. Velocity distribution around Airfoil-at AOA=6° and NACA 63(4)-221 at AOA=6° and v=10m/s

Fig. 18.Velocity vector distribution around Airfoil-NACA 63(4)-221 AOA=6° and v=9 m/s

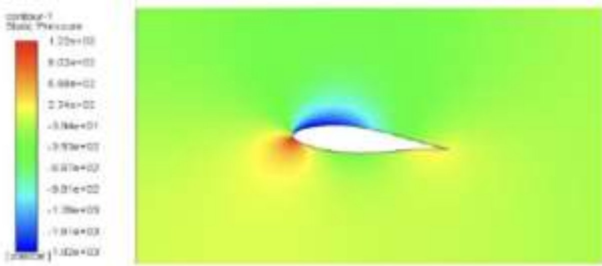


Fig. 22. Pressure distribution around Airfoil-at NACA 63(4)-221 at AOA=6° and v =43.79 m/s

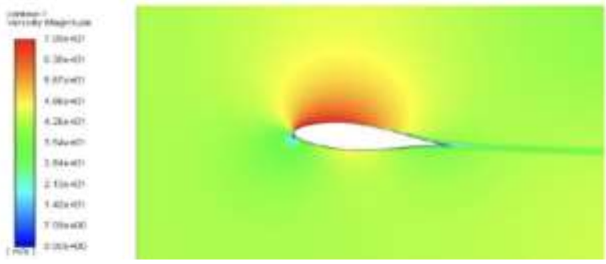


Fig. 18.Velocity distribution around Airfoil-NACA 63(4)-221 at AOA=6° and v= 43.79m/s

Fig. 22. Velocity vector distribution around Airfoil- NACA 63(4)-221 at AOA=6° and v =87.58 m/s

Fig. 18. Velocity vector distribution around Airfoil-NACA 63(4)-221 AOA=6° and v= 43.79m/s

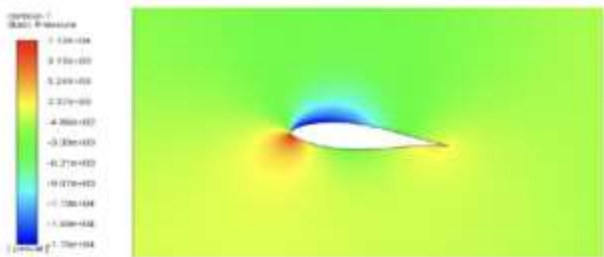


Fig. 22. Pressure distribution around Airfoil-at NACA 63(4)-221 at AOA=6° and v = 131.37 m/s

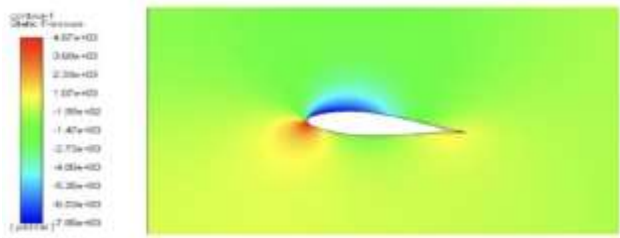


Fig. 18. Velocity distribution around Airfoil-NACA 63(4)-221 and v= 43.79m/s

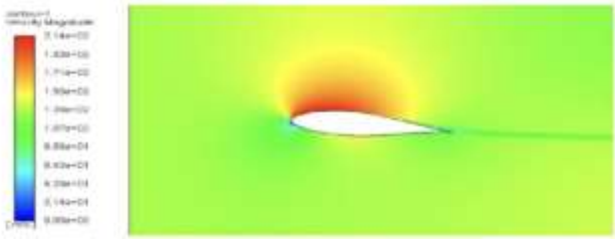


Fig. 22. Velocity distribution around Airfoil-at AOA=6° NACA 63(4)-221 at AOA=6° and v = 131.37 m/s

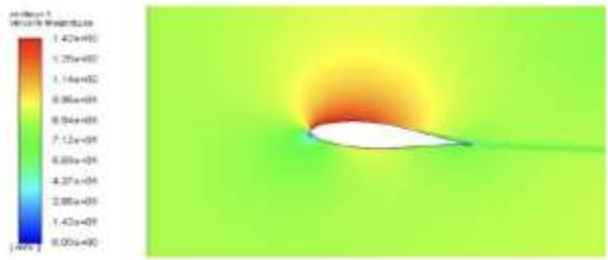


Fig. 18.Velocity distribution around Airfoil-NACA 63(4)-221 at AOA=6° and v= 487.58 m/s

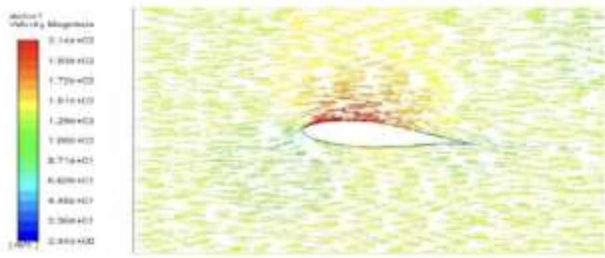


Fig. 22. Velocity vector distribution around Airfoil-NACA 63(4)-221 at AOA=6° and v =131.37 m/s

**Table VI. CFD analysis results of NACA airfoil 63(4)-221 at various angle of attack and wind speed for low Reynolds number ( x 105)**

| Angle of attack (Degree) | Reynolds No. | Wind speed (m/s) | Drag force (N) | Drag Coefficient, $C_d$ | Corrected drag coefficient, $C_d$ | Lift force (N) | Lift coefficient, $C_l$ | $C_l/C_d$ ratio |
|---|---|---|---|---|---|---|---|---|
| 4° | 397061.74 | 6.0 | 0.48741 | 0.02210 | 0.00884 | 13.81279 | 0.62642 | 70.84730 |
| 4° | 463238.70 | 7.0 | 0.63990 | 0.02132 | 0.00852 | 18.99529 | 0.63290 | 74.21159 |
| 4° | 529415.66 | 8.0 | 0.81047 | 0.02067 | 0.00827 | 24.99530 | 0.63763 | 77.10047 |
| 4° | 595592.61 | 9.0 | 1.00014 | 0.02015 | 0.00806 | 31.82257 | 0.64141 | 79.54481 |
| 4° | 661769.57 | 10.0 | 1.21105 | 0.01977 | 0.00790 | 39.45804 | 0.64420 | 81.45355 |
| 5° | 397061.74 | 6.0 | 0.53559 | 0.02428 | 0.00971 | 16.24221 | 0.73659 | 75.81421 |
| 5° | 463238.70 | 7.0 | 0.70406 | 0.02345 | 0.00938 | 22.32535 | 0.74386 | 79.27356 |
| 5° | 529415.66 | 8.0 | 0.89296 | 0.02277 | 0.00911 | 29.39745 | 0.74993 | 82.30328 |
| 5° | 595592.61 | 9.0 | 1.10418 | 0.02225 | 0.00890 | 37.43394 | 0.75451 | 84.75457 |
| 5° | 661769.57 | 10.0 | 1.33887 | 0.02185 | 0.00874 | 46.42601 | 0.75797 | 86.68825 |
| 6° | 397061.74 | 6.0 | 0.59475 | 0.02697 | 0.01078 | 18.66507 | 0.84647 | 78.45724 |
| 6° | 463238.70 | 7.0 | 0.78231 | 0.02606 | 0.01042 | 25.70568 | 0.85649 | 82.14612 |
| 6° | 529415.66 | 8.0 | 0.99446 | 0.02536 | 0.01014 | 33.87317 | 0.86410 | 85.15458 |
| 6° | 595592.61 | 9.0 | 1.23138 | 0.02481 | 0.00992 | 43.16754 | 0.87008 | 87.64042 |
| 6° | 661769.57 | 10.0 | 1.49685 | 0.02443 | 0.00977 | 53.49336 | 0.87335 | 89.34293 |
| 7° | 397061.74 | 6.0 | 0.66733 | 0.03026 | 0.01210 | 20.82543 | 0.94445 | 78.01681 |
| 7° | 463238.70 | 7.0 | 0.87819 | 0.02926 | 0.01170 | 28.72091 | 0.95695 | 81.76109 |
| 7° | 529415.66 | 8.0 | 1.11739 | 0.02850 | 0.01140 | 37.88486 | 0.96644 | 84.76180 |
| 7° | 595592.61 | 9.0 | 1.38664 | 0.02794 | 0.01117 | 48.29977 | 0.97352 | 87.08039 |
| 7° | 661769.57 | 10.0 | 1.68755 | 0.02755 | 0.01102 | 59.93634 | 0.97854 | 88.79193 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 8° | 397061.74 | 6.0 | 0.75943 | 0.03444 | 0.01377 | 22.68052 | 1.02858 | 74.66238 |
| 8° | 463238.70 | 7.0 | 0.99889 | 0.03328 | 0.01331 | 31.37178 | 1.04528 | 78.51610 |
| 8° | 529415.66 | 8.0 | 1.27136 | 0.03243 | 0.01297 | 41.47370 | 1.05799 | 81.55338 |
| 8° | 595592.61 | 9.0 | 1.57887 | 0.03182 | 0.01272 | 52.94724 | 1.06720 | 83.83720 |
| 8° | 661769.57 | 10.0 | 1.92250 | 0.03138 | 0.01255 | 65.77964 | 1.07394 | 85.53898 |

**Table VII. CFD analysis results of NACA airfoil 63(4)-221 at various angle of attack and wind speed for high Reynolds number ( x 106)**

| Angle of attack | Reynolds No. | Wind speed (m/s) | Drag force (N) | Drag Coefficient, $C_d$ | Corrected drag coefficient, $C_d$ | Lift force (N) | Lift coefficient, $C_l$ | $C_l/C_d$ ratio |
|---|---|---|---|---|---|---|---|---|
| 5° | 3000000 | 43.79 | 21.35365 | 0.01817 | 0.00727 | 929.77072 | 0.79151 | 108.85381 |
| 5° | 6000000 | 87.58 | 78.92014 | 0.01679 | 0.00671 | 3780.4004 | 0.80456 | 119.75397 |
| 5° | 9000000 | 131.37 | 170.4888 | 0.01612 | 0.00645 | 8579.2665 | 0.81149 | 125.80395 |
| 6° | 3000000 | 43.79 | 24.08764 | 0.02050 | 0.00820 | 1075.4018 | 0.91548 | 111.61343 |
| 6° | 6000000 | 87.58 | 89.59506 | 0.01906 | 0.00762 | 4374.3581 | 0.93096 | 122.05913 |
| 6° | 9000000 | 131.37 | 194.0922 | 0.01835 | 0.00734 | 9920.3236 | 0.93834 | 127.77846 |
| 7° | 3000000 | 43.79 | 27.30909 | 0.02324 | 0.00929 | 1213.8412 | 1.03333 | 111.12058 |
| 7° | 6000000 | 87.58 | 102.00160 | 0.02170 | 0.00868 | 4943.4744 | 1.05209 | 121.16169 |
| 7° | 9000000 | 131.37 | 221.81895 | 0.02098 | 0.00839 | 11218.193 | 1.06111 | 126.43410 |
| 8° | 3000000 | 43.79 | 31.22440 | 0.02658 | 0.01063 | 1352.3971 | 1.15129 | 108.28045 |
| 8° | 6000000 | 87.58 | 117.45439 | 0.02499 | 0.00999 | 5528.5577 | 1.17661 | 117.67456 |
| 8° | 9000000 | 131.37 | 255.76350 | 0.02419 | 0.00967 | 12557.665 | 1.18780 | 122.74684 |

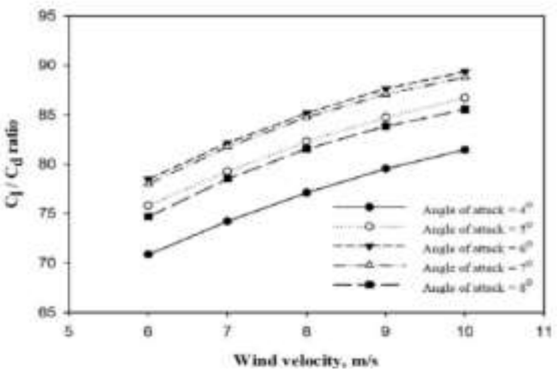



Fig. 12. Cl/Cd ratio at different angle of attack for different wind velocity at low Reynolds number (x 105)

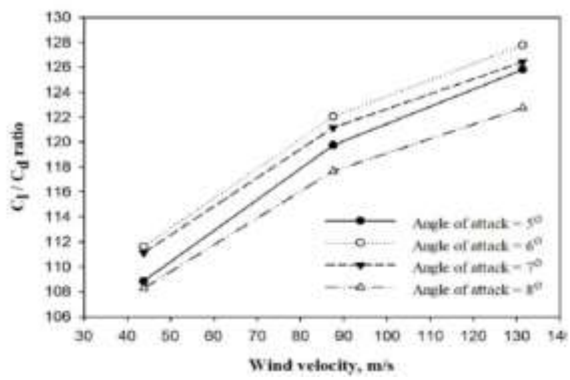



Fig. 13. Cl/Cd ratio at different angle of attack for different wind velocity at high Reynolds number (x 106)

VI. RESULTS

A. Pressure Distribution around Airfoil

The static pressure contour is shown in Fig. 7, 10, 13, 16, 19, 22, 25 and 28 for angle of attack 6 degree, because at this angle of attack Cl/Cd ratio is maximum as given Table VI and VII. The pressure at the bottom surface of airfoil for incoming flow is more than upper surface so the incoming air can effectively push the airfoil upward normal to flow direction of air.

B. Velocity Distribution around Airfoil

The Velocity distribution contour is shown in Fig. 8, 11, 14, 17, 20, 23, 26 and 29 for angle of attack 6 deg.

C. Velocity Vector around Airfoil

The Velocity vector contour is shown in Fig. 9, 12, 15, 18, 21, 24, 27 and 30 for angle of attack 6 deg.

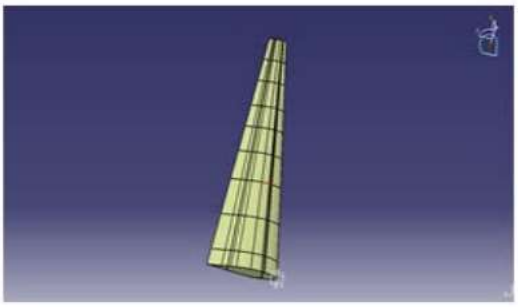
D. Comparison of Simulation and Experimental Results

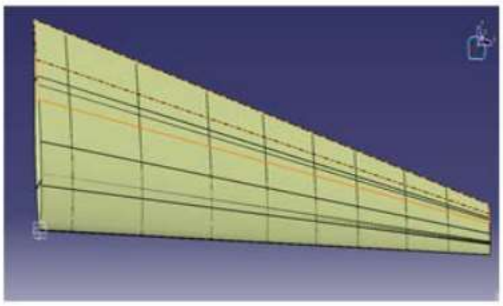
There are two forces and one moment works on an airfoil. The force component which is normal to the incoming flow stream is known as lift force and the component which is acting parallel to the flow stream is known as drag force. In this analysis, first of all simulation is carried out and then results are being verified with results available in previous literature. Here simulation is done for angle of attack 4° to 8°. Results shows the ratio lift coefficient to drag coefficient (Cl/Cd) increases with increasing angle of attack from 4° to 6° and then again decreasing with further increasing AOA. Hence AOA=6° shows maximum Cl/Cd ratio for both condition of low and high Reynolds number. This can be further use in blade design process. It is also find out that for any fix value of AOA, the Cl/Cd ratio increases with increasing wind velocity or increasing Reynolds. Lift coefficient also increases with increasing angle of attack 4° to 8° and Reynolds number, while drag coefficient decreases.

## VI. Static Load Test of Wing Structure and Loads Using FEM

Analysis are conducted mainly, normal mode and linear static analyses. The wing geometry is imported from Fusion 360, figure (1), into ANSYS®2022 by extracting each points and curves for geometrical accuracy of the model. In the wing model structure there are eight ribs and two spars. The types of mesh, used for the spars, ribs and skin ($3mm$) are as in table.

| Structure | Element |
|---|---|
| Spars | 3D Tetrahedral |
| Ribs | 3D Tetrahedral |
| Skin | 2D Quad lateral |





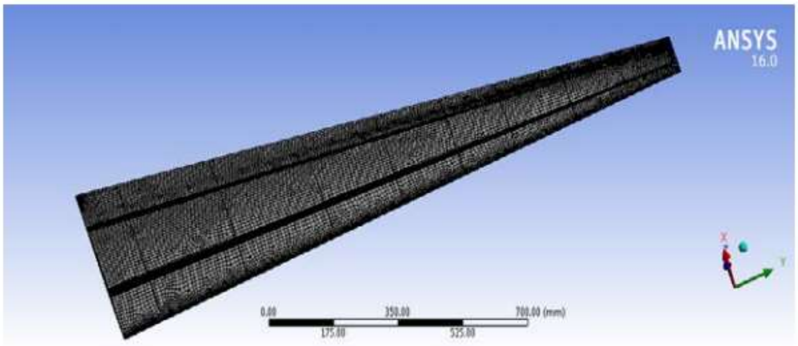
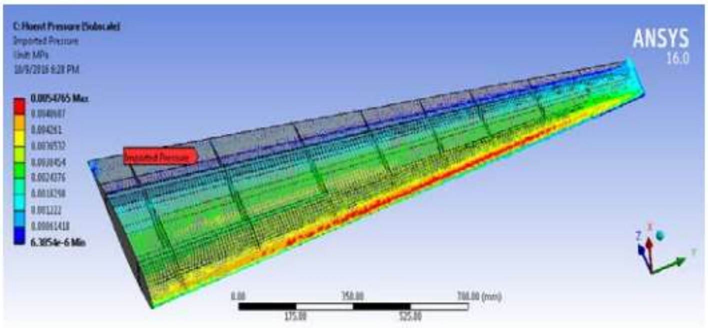


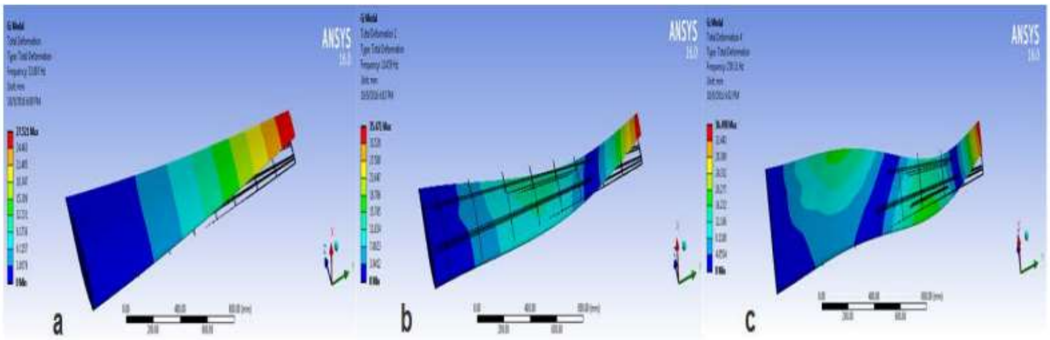

Figure above shows the distribution and the location of the ribs (8 ribs) along the wing span located at different distances. The wing structure has two spars, the front spar is the I section located at 21% of the root chord, and the rear spar is the C section located at 70% of root chord, as seen in figure. The wing skin is modeled as in figure below, the material is aluminum alloy 7068 and the thickness is 3 mm.

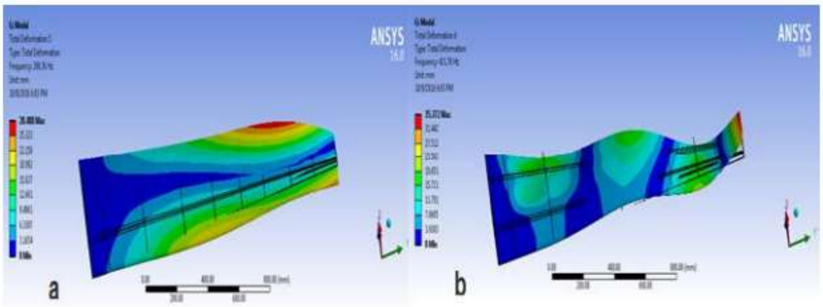

Figure below shows the imported load from FLUENT; as shown, the maximum pressure in the leading edge is $5 * 10^{-3}$ Mpa, and the minimum at the trailing edge is $6.385 * 10^{-6}$ Mpa.




RESULTS AND DISCUSSIONS

a. Normal mode analysis: Finite element analysis (FEA) approach was used in term of normal mode analysis in order to obtain both modal properties of the wing structure which are the natural frequencies and the mode shape. Figure below shows the wing model bending modes; a- 1st bending mode with frequency33.007 Hz, b- 2nd bending mode with frequency 114.59 Hz, c- 3rd bending mode with frequency250.31 Hz. Figure below shows the twisting and combined bending and twisting mode with frequency421.78 Hz.



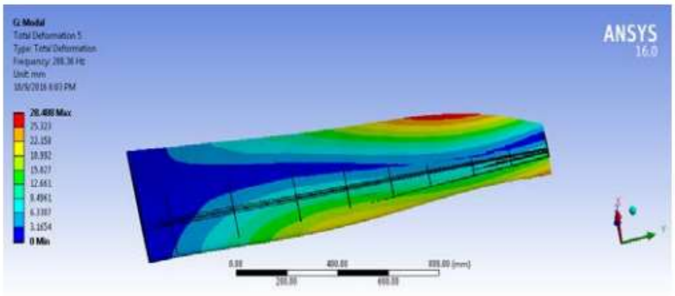




Figure 7: Twisting mode

| No | Mode | Frequency (Hz) |
|---|---|---|
| 1 | First bending | 33.007 |
| 2 | Second bending | 114.59 |
| 3 | Third bending | 250.31 |
| 4 | Twisting | 288.36 |
| 5 | Combined bending and twitting | 421.78 |

b. Linear static structure: The purpose of linear static structure is to find stresses, strains, deformation and safety factor of the wing at effected loads. As shown in figure below, the maximum total deformation in tip is 26.922 $mm$, and there is no deformation at wing root, because the wing is acting as a cantilever beam fixed at the root and free on its tip.

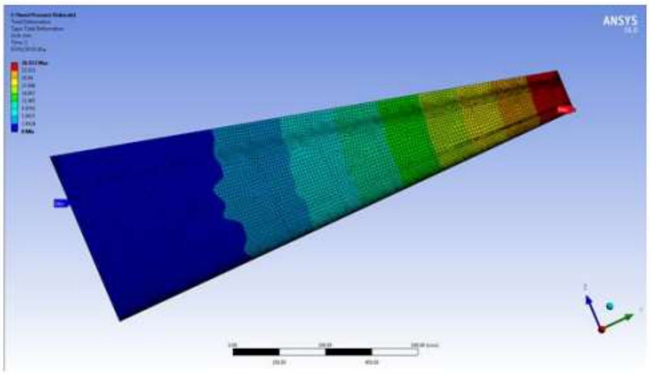
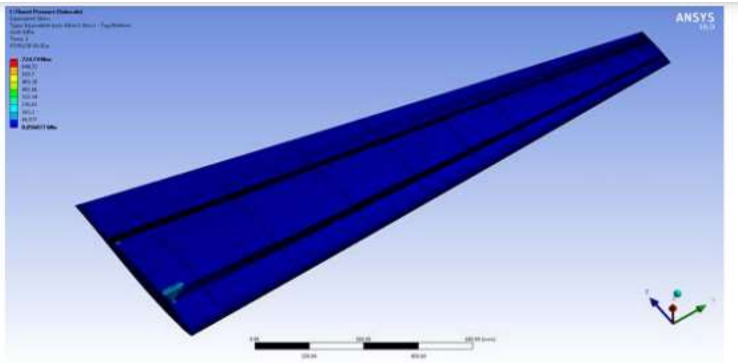
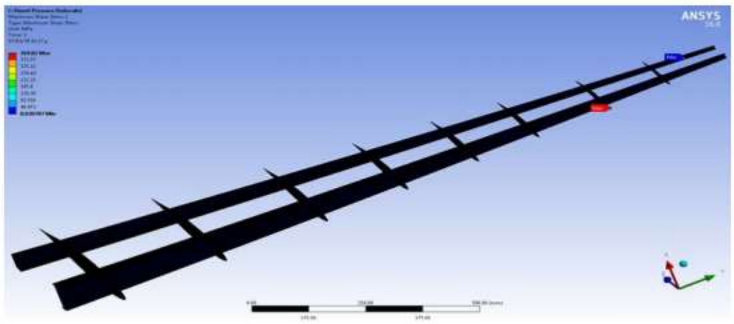



Figure below shows the Von-Mises stress; the maximum Von-Mises Stress is 724.74 Mpa, and the minimum is 0.057 Mpa. Figure below shows that the Maximum shear stress in an interior component (spars &ribs) is 418.02 Mpa, and the minimum is 0.030 Mpa.




As shown in figure (11), the maximum equivalent elastic strain is 0.01 $mm/mm$, and the minimum is $1.461 * 10^{-6}$ $mm/mm$.

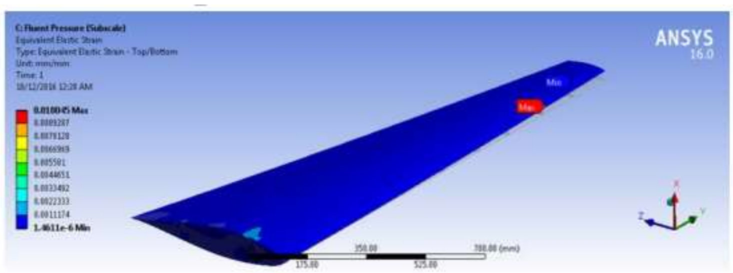



Figure 11: Equivalent Elastic Strain

CONCLUSIONS

An aircraft wing model is analyzed for to determine the normal mode analysis and the linear static structure. The finite element analysis (FEA) approach was used for obtaining the modal properties of the wing structure which are the natural frequencies and the mode shape. The linear static structure is conducted to find stresses, strains, deformation and safety factor of the wing at various loadings.

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[1] M. C.-Y. Niu, Airframe Structural Design: Practical Design Information, Second edition, Hong Kong, 1999.

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# 5. Experimental approach

Determining wing lift and drag using load cells involves measuring the forces directly acting on the wing. Load cells are sensors designed to measure forces, and they can be utilized in experimental setups to obtain lift and drag data. Here's an outline of an experimental approach using load cells:

## Subject wing :

Fig. 5.1.

**Refrence wing:**

Fig. 5.2

# Setup:

1. **Instrumentation:** Attach load cells at fuselage aft on the wing model to directly measure the forces acting on the wing. Load cells can be placed at mounting onto a test rig that simulates the airflow conditions.

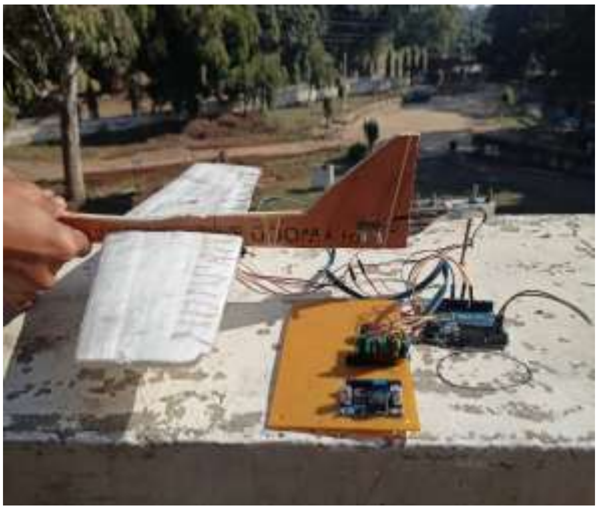

Fig 5.1. All DAC assembly

Fig.5.2 . Mounting

2. **Calibration:** Before the experiment, calibrate the load cells to ensure accurate force measurements. This calibration involves applying known forces to the load cell and correlating the output signal to the applied force. The procedure is as follow as:

- Remove all loads from load cell
- Then press ' t ' onto output terminal which tare the instrument.
- Then hang the known weight load on load cell.
- Then enter the value of weight onto output terminal

3. **DAS( Data Acquisition System):** The load cell is paired with hx711 and ESP32 for to work as DAS . Load cell is acted as force and moment sensor . HX711 is working as amplifier and ADC . And ESP32 as microcontroller which transmitters the data wirelessly to observing computer.

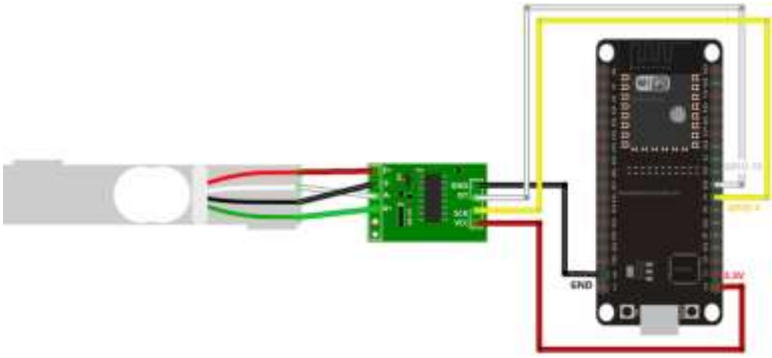

Fig.5.3. DAS circuit

## {Code}

The algorithim is coded onto Arduino IDE 2.2.2

```
#include "HX711.h"
#include <LiquidCrystal.h>


const int LOADCELL_DOUT_PIN = 2;
const int LOADCELL_SCK_PIN = 3;


HX711 scale;

double mtime;
double force;
double dt;
double dp;
double impulse;
double last;
double ig;

void setup() {

  Serial.begin(115200);
```

```
  // Serial.println("Take off Calibration Weight!");


  //not my code

  int done = 1;  // Declared as a global

  pinMode(7,OUTPUT);

  while (Serial.available() > 0) {
    byte dummyread = Serial.read();
  }

  //back to me

  scale.begin(LOADCELL_DOUT_PIN, LOADCELL_SCK_PIN);

  double taret = 80;
  scale.tare(taret);
  scale.set_scale(194.25);  // paste cal val here

}


void loop() {
  digitalWrite(7,HIGH);
  mtime = micros() / 1000000.f;
  force = scale.get_units() * 0.009806f;
  dt = mtime - last;
  dp = force * dt;
  impulse = impulse + dp;
  last = mtime;
  Serial.print("Thrust:");
  Serial.print(force, 10);
  Serial.print("\n");
  delay(1);
}
```

### Experimental Procedure:

1. **Mounting and Preparation**: Install the wing model on the load that is place over test rig . On the DAS( Data Acquisition System) as by powering the Arduino. The wing is mounted to the test rig via load cell which measures the moment as shown in fig.5.2

2.  **Data Collection:**  Apply a range of airflow conditions, including varying airspeeds and angles of attack, while recording the data from the load cells. The load cells will measure the forces acting on the wing, including lift and drag. Velocity ranges from 16 m/s – 94 m/s. while angle ranges from +75 degree to  -70 degree.The angle is varied manually.

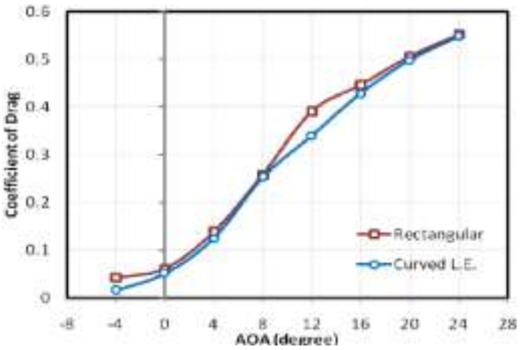


Fig .5.4. cd vs aoa plot

3.  **Force Calculation:**  Analyze the data from the load cells to separate the forces into their respective components—lift and drag. The angle at which the load cell is mounted and the orientation of the wing model will help distinguish between the forces.

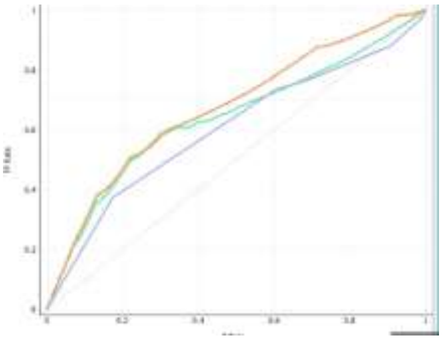

Fig.5.5 lift vs velocity

4.  **Variation Testing:**  Systematically vary parameters like angle of attack, wing geometry, airspeed, and surface properties to observe their effects on lift and drag. This allows for the determination of how these factors influence the aerodynamic performance of the wing.

5.  **Repeated Trials**:  Conduct multiple trials to ensure consistency and reliability of the data. This helps in reducing experimental errors and confirms the repeatability of the results.

## Considerations:

1. **Environmental Conditions**:  Maintain controlled and consistent environmental conditions throughout the experiments to ensure accurate and comparable data.

2. **Data Analysis**: Accurately process and analyze the data obtained from the load cells to extract meaningful information about lift and drag forces. The data points is imported to excel and then ploted with AOA.

3. **Safety Measures**: Ensure safety protocols are followed, especially if conducting experiments in wind tunnels or with high-speed airflow setups.

Utilizing load cells in experimental setups offers a direct method of measuring aerodynamic forces on a wing, providing valuable data for understanding its lift and drag characteristics.

# 8. Future Aspect

The future prospects of closed-wing fighter aircraft involve a combination of aerodynamic considerations, advanced technologies, and evolving mission requirements. Here are several potential aspects to consider:

1. Reduced Induced Drag:

Closed-wing configurations can contribute to reduced induced drag, enhancing the overall aerodynamic efficiency of fighter aircraft. This could result in improved fuel efficiency, extended operational range, and enhanced mission endurance.

2. Enhanced Maneuverability:

The closed-wing design, with its potential for increased structural rigidity and efficient lift distribution, may offer superior maneuverability. Fighter aircraft equipped with closed-wing configurations could excel in dogfighting scenarios, providing pilots with a more agile and responsive platform.

3. Stealth and Low Observability:

Closed-wing designs could be explored for their potential contributions to stealth and low observability. The interconnected wings may facilitate smoother contours and reduced radar cross-section, enhancing the aircraft's ability to operate in contested environments with minimized detection.

4. Innovative Control Surfaces:

The closed-wing design allows for innovative control surface arrangements. This could lead to the development of advanced control systems that exploit the unique aerodynamic characteristics of the closed-wing configuration, providing pilots with more precise and effective control over the aircraft.

5. Integration with Advanced Avionics:

Future closed-wing fighter aircraft are likely to incorporate advanced avionics and sensor technologies. These could include next-generation radar systems, sensor fusion capabilities, and artificial intelligence to enhance situational awareness, target acquisition, and overall mission effectiveness.

6. Adaptability to New Propulsion Systems:

Closed-wing designs may be well-suited for integration with advanced propulsion systems, such as electric or hybrid-electric propulsion. This could provide improved energy efficiency, reduced reliance on traditional fuels, and the potential for enhanced performance characteristics.

7. Versatility in Mission Profiles:

Closed-wing fighter aircraft may be designed to excel in a variety of mission profiles, from air superiority and interception to ground attack and electronic warfare. The adaptability of the closed-wing configuration could make these aircraft versatile assets for a range of military operations.

8. Human-Machine Interface Enhancements:

Future closed-wing fighters are likely to incorporate advanced human-machine interface technologies. This could include augmented reality displays, intuitive control systems, and enhanced pilot-vehicle interaction to optimize the pilot's ability to handle the complexities of the aircraft in dynamic and high-stress situations.

9. Autonomous and Swarm Capabilities:

Closed-wing fighter aircraft may be designed to operate in conjunction with autonomous systems and swarm tactics. The interconnected wings could facilitate coordination among multiple aircraft, enabling collaborative and synchronized mission execution.

10. Challenges and Trade-offs:

Engineers will need to address challenges related to weight, complexity, and control system integration. Striking the right balance between aerodynamic performance, structural integrity, and mission requirements will be crucial for the success of closed-wing fighter aircraft.

As with any speculative analysis, these future aspects are contingent on continued advancements in aerospace technology, evolving military strategies, and the specific needs of defense forces worldwide. Ongoing research and development will shape the trajectory of closed-wing fighter aircraft in the years to come.